



Datenbanken

6.6 | SQL - Daten verwalten (1v2)

Gliederung des Vorlesungsabschnitts zur SQL

- Überblick
- Datenbanken verwalten
- Tabellen verwalten
- Datentypen
- Daten verwalten
- Einfache Datenbankabfragen
- Funktionen in Datenbankabfragen
- Datenbankabfragen über mehrere Tabellen
- Subqueries
- Sichten
- Locking und Transaktionsverwaltung



SQL

Daten verwalten



Überblick

- Es gibt zahlreiche Befehle zum Verarbeiten von Daten:

Befehle zur Datenverarbeitung

INSERT

Daten manuell einfügen

LOAD DATA INFILE

Daten aus einer Datei importieren

UPDATE

Daten aktualisieren

DELETE

Daten löschen



Daten einfügen: INSERT

```
INSERT [IGNORE]
INTO tbl_name [(col_name,...)]
VALUES ({expr | DEFAULT},...), (...), ...
[ ON DUPLICATE KEY UPDATE col_name=expr, ... ]
```

- Mit dem **INSERT** Befehl können Daten manuell in einer Tabelle erfasst werden
- **tbl_name** nimmt hierbei den Tabellennamen auf
- **expr** steht für die einzutragenden Daten
- mehrere Datensätze können gleichzeitig eingefügt werden

Daten einfügen: INSERT

```
mysql> CREATE TABLE person(  
    -> person_id INT PRIMARY KEY AUTO_INCREMENT,  
    -> name CHAR(35),  
    -> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)
```

Daten einfügen: INSERT

```
mysql> CREATE TABLE person(  
    -> person_id INT PRIMARY KEY AUTO_INCREMENT,  
    -> name CHAR(35),  
    -> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> INSERT INTO person VALUES (1, 'Skywalker', 'Anakin');  
Query OK, 1 row affected (0.00 sec)
```

Daten einfügen: INSERT

```
mysql> CREATE TABLE person(  
    -> person_id INT PRIMARY KEY AUTO_INCREMENT,  
    -> name CHAR(35),  
    -> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> INSERT INTO person VALUES (1, 'Skywalker', 'Anakin');  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO person VALUES ('Skywalker', 'Luke');  
ERROR 1136 (21S01): Column count doesn't match value count at row 1  
mysql>
```




Daten einfügen: INSERT

- Das Datenformat der eingefügten Werte muss zur Tabellendefinition passen
- Dennoch müssen nicht alle Felder mit Werten befüllt werden:
 - **NULL** Das Datenfeld bleibt leer (null).
 - **DEFAULT** Es wird der **DEFAULT** Wert eingefügt
 - **0** bei **AUTO_INCREMENT** wird der nächste sinnvolle Wert genommen

```
mysql> CREATE TABLE person(  
    -> person_id INT PRIMARY KEY AUTO_INCREMENT,  
    -> name CHAR(35) DEFAULT 'Skywalker',  
    -> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE TABLE person(  
    -> person_id INT PRIMARY KEY AUTO_INCREMENT,  
    -> name CHAR(35) DEFAULT 'Skywalker',  
    -> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> INSERT INTO person VALUES (0, 'Skywalker', 'Anakin');  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> CREATE TABLE person(  
    -> person_id INT PRIMARY KEY AUTO_INCREMENT,  
    -> name CHAR(35) DEFAULT 'Skywalker',  
    -> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> INSERT INTO person VALUES (0, 'Skywalker', 'Anakin');  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO person VALUES (0, DEFAULT, 'Luke' );  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> CREATE TABLE person(  
    -> person_id INT PRIMARY KEY AUTO_INCREMENT,  
    -> name CHAR(35) DEFAULT 'Skywalker',  
    -> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> INSERT INTO person VALUES (0, 'Skywalker', 'Anakin');  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO person VALUES (0, DEFAULT, 'Luke' );  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO person VALUES (0, NULL, 'Yoda' );  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> CREATE TABLE person(  
    -> person_id INT PRIMARY KEY AUTO_INCREMENT,  
    -> name CHAR(35) DEFAULT 'Skywalker',  
    -> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> INSERT INTO person VALUES (0, 'Skywalker', 'Anakin');  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO person VALUES (0, DEFAULT, 'Luke' );  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO person VALUES (0, NULL, 'Yoda' );  
Query OK, 1 row affected (0.01 sec)  
  
mysql> SELECT * FROM person;  
+-----+-----+-----+  
| person_id | name      | vorname |  
+-----+-----+-----+  
|          1 | Skywalker | Anakin  |  
|          2 | Skywalker | Luke    |  
|          3 | NULL      | Yoda    |  
+-----+-----+-----+  
3 rows in set (0.00 sec)  
  
mysql>
```



Daten einfügen: INSERT

```
INSERT [IGNORE]
INTO tbl_name [(col_name,...)]
VALUES ({expr | DEFAULT},...), (...), ...
[ ON DUPLICATE KEY UPDATE col_name=expr, ... ]
```

- Daten können durch die zusätzliche Angabe von Spaltennamen gezielt in ausgewählte Spalten eingefügt werden.
- Die unberücksichtigten Spalten werden mit Ihren **DEFAULT** Werten oder mit **NULL** initialisiert

Daten einfügen: INSERT

```
mysql> CREATE TABLE skywalkers (person_id INT PRIMARY KEY AUTO_INCREMENT, name CHAR(35)  
DEFAULT 'Skywalker', vorname CHAR(35), geburtsdatum DATE);  
Query OK, 0 rows affected (0.01 sec)
```


Daten einfügen: INSERT

```
mysql> CREATE TABLE skywalkers (person_id INT PRIMARY KEY AUTO_INCREMENT, name CHAR(35)
DEFAULT 'Skywalker', vorname CHAR(35), geburtsdatum DATE);
Query OK, 0 rows affected (0.01 sec)

mysql> INSERT INTO skywalkers(vorname) VALUES ('Anakin'),('Luke'),('Padmé');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

Daten einfügen: INSERT

```
mysql> CREATE TABLE skywalkers (person_id INT PRIMARY KEY AUTO_INCREMENT, name CHAR(35)
DEFAULT 'Skywalker', vorname CHAR(35), geburtsdatum DATE);
Query OK, 0 rows affected (0.01 sec)

mysql> INSERT INTO skywalkers(vorname) VALUES ('Anakin'),('Luke'),('Padmé');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM skywalkers;
+-----+-----+-----+-----+
| person_id | name      | vorname | geburtsdatum |
+-----+-----+-----+-----+
|          1 | Skywalker | Anakin  | NULL         |
|          2 | Skywalker | Luke    | NULL         |
|          3 | Skywalker | Padmé   | NULL         |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

```
mysql> CREATE TABLE skywalkers (person_id INT PRIMARY KEY AUTO_INCREMENT, name CHAR(35)  
DEFAULT 'Skywalker', vorname CHAR(35), geburtsdatum DATE NOT NULL);  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE TABLE skywalkers (person_id INT PRIMARY KEY AUTO_INCREMENT, name CHAR(35)  
DEFAULT 'Skywalker', vorname CHAR(35), geburtsdatum DATE NOT NULL);  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> INSERT INTO skywalkers(vorname) VALUES ('Anakin'),('Luke'),('Padmé');  
ERROR 1364 (HY000): Field 'geburtsdatum' doesn't have a default value
```

```
mysql> CREATE TABLE skywalkers (person_id INT PRIMARY KEY AUTO_INCREMENT, name CHAR(35)
DEFAULT 'Skywalker', vorname CHAR(35), geburtsdatum DATE NOT NULL);
Query OK, 0 rows affected (0.01 sec)

mysql> INSERT INTO skywalkers(vorname) VALUES ('Anakin'),('Luke'),('Padmé');
ERROR 1364 (HY000): Field 'geburtsdatum' doesn't have a default value
mysql> INSERT INTO skywalkers(vorname) VALUES ('Anakin', '2723-11-14'),('Luke','2755-12-
09' ),('Padmé','2728-07-02');
ERROR 1136 (21S01): Column count doesn't match value count at row 1
```

```
mysql> CREATE TABLE skywalkers (person_id INT PRIMARY KEY AUTO_INCREMENT, name CHAR(35)
DEFAULT 'Skywalker', vorname CHAR(35), geburtsdatum DATE NOT NULL);
Query OK, 0 rows affected (0.01 sec)

mysql> INSERT INTO skywalkers(vorname) VALUES ('Anakin'),('Luke'),('Padmé');
ERROR 1364 (HY000): Field 'geburtsdatum' doesn't have a default value
mysql> INSERT INTO skywalkers(vorname) VALUES ('Anakin', '2723-11-14'),('Luke', '2755-12-
09' ),('Padmé', '2728-07-02');
ERROR 1136 (21S01): Column count doesn't match value count at row 1
mysql> INSERT INTO skywalkers(vorname, geburtsdatum) VALUES ('Anakin', '2723-11-14'),('L
uke', '2755-12-09' ),('Padmé', '2728-07-02');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> CREATE TABLE skywalkers (person_id INT PRIMARY KEY AUTO_INCREMENT, name CHAR(35)
DEFAULT 'Skywalker', vorname CHAR(35), geburtsdatum DATE NOT NULL);
Query OK, 0 rows affected (0.01 sec)

mysql> INSERT INTO skywalkers(vorname) VALUES ('Anakin'),('Luke'),('Padmé');
ERROR 1364 (HY000): Field 'geburtsdatum' doesn't have a default value
mysql> INSERT INTO skywalkers(vorname) VALUES ('Anakin', '2723-11-14'),('Luke', '2755-12-
09' ),('Padmé', '2728-07-02');
ERROR 1136 (21S01): Column count doesn't match value count at row 1
mysql> INSERT INTO skywalkers(vorname, geburtsdatum) VALUES ('Anakin', '2723-11-14'),('L
uke', '2755-12-09' ),('Padmé', '2728-07-02');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM skywalkers;
+-----+-----+-----+-----+
| person_id | name      | vorname | geburtsdatum |
+-----+-----+-----+-----+
|          1 | Skywalker | Anakin  | 2723-11-14   |
|          2 | Skywalker | Luke    | 2755-12-09   |
|          3 | Skywalker | Padmé   | 2728-07-02   |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```



Daten einfügen: INSERT

```
INSERT [IGNORE]
      INTO tbl_name [(col_name,...)]
      VALUES ({expr | DEFAULT},...), (...), ...
      [ ON DUPLICATE KEY UPDATE col_name=expr, ... ]
```

- **IGNORE** führt dazu, dass ein Fehler bei einer **INSERT**-Operationen ignoriert wird:
 - es wird lediglich eine Warnung ausgegeben und die Einfügeoperation übersprungen
 - das restliche Programm läuft weiter
- sonst würde das Programm bei einem Fehler abbrechen


```
mysql> CREATE TABLE personen (person_id INT PRIMARY KEY AUTO_INCREMENT,  
-> name CHAR(35),  
-> vorname CHAR(35),  
-> geburtsdatum DATE);  
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> CREATE TABLE personen (person_id INT PRIMARY KEY AUTO_INCREMENT,  
-> name CHAR(35),  
-> vorname CHAR(35),  
-> geburtsdatum DATE);  
Query OK, 0 rows affected (0.05 sec)  
  
mysql> INSERT INTO personen VALUES (1, 'Skywalker', 'Anakin', NULL);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> CREATE TABLE personen (person_id INT PRIMARY KEY AUTO_INCREMENT,  
-> name CHAR(35),  
-> vorname CHAR(35),  
-> geburtsdatum DATE);  
Query OK, 0 rows affected (0.05 sec)  
  
mysql> INSERT INTO personen VALUES (1, 'Skywalker', 'Anakin', NULL);  
Query OK, 1 row affected (0.01 sec)  
  
mysql> INSERT INTO personen VALUES (1, 'Skywalker', 'LUKE', NULL), (2, 'Solo', 'Ben', NULL);  
ERROR 1062 (23000): Duplicate entry '1' for key 'PRIMARY'
```

```
mysql> CREATE TABLE personen (person_id INT PRIMARY KEY AUTO_INCREMENT,  
-> name CHAR(35),  
-> vorname CHAR(35),  
-> geburtsdatum DATE);  
Query OK, 0 rows affected (0.05 sec)  
  
mysql> INSERT INTO personen VALUES (1, 'Skywalker', 'Anakin', NULL);  
Query OK, 1 row affected (0.01 sec)  
  
mysql> INSERT INTO personen VALUES (1, 'Skywalker', 'LUKE', NULL), (2, 'Solo', 'Ben', NULL);  
ERROR 1062 (23000): Duplicate entry '1' for key 'PRIMARY'  
mysql> SELECT * FROM personen;  
+-----+-----+-----+-----+  
| person_id | name       | vorname | geburtsdatum |  
+-----+-----+-----+-----+  
|          1 | Skywalker | Anakin  | NULL         |  
+-----+-----+-----+-----+  
1 row in set (0.00 sec)
```

```
mysql> CREATE TABLE personen (person_id INT PRIMARY KEY AUTO_INCREMENT,  
-> name CHAR(35),  
-> vorname CHAR(35),  
-> geburtsdatum DATE);  
Query OK, 0 rows affected (0.05 sec)  
  
mysql> INSERT INTO personen VALUES (1, 'Skywalker', 'Anakin', NULL);  
Query OK, 1 row affected (0.01 sec)  
  
mysql> INSERT INTO personen VALUES (1, 'Skywalker', 'LUKE', NULL), (2, 'Solo', 'Ben', NULL);  
ERROR 1062 (23000): Duplicate entry '1' for key 'PRIMARY'  
mysql> SELECT * FROM personen;  
+-----+-----+-----+-----+  
| person_id | name       | vorname | geburtsdatum |  
+-----+-----+-----+-----+  
|          1 | Skywalker | Anakin  | NULL         |  
+-----+-----+-----+-----+  
1 row in set (0.00 sec)  
  
mysql> INSERT IGNORE INTO personen VALUES (1, 'Skywalker', 'LUKE', NULL), (2, 'Solo', 'Ben', NULL);  
Query OK, 1 row affected, 1 warning (0.00 sec)  
Records: 2  Duplicates: 1  Warnings: 1
```

```
mysql> CREATE TABLE personen (person_id INT PRIMARY KEY AUTO_INCREMENT,  
-> name CHAR(35),  
-> vorname CHAR(35),  
-> geburtsdatum DATE);  
Query OK, 0 rows affected (0.05 sec)  
  
mysql> INSERT INTO personen VALUES (1, 'Skywalker', 'Anakin', NULL);  
Query OK, 1 row affected (0.01 sec)  
  
mysql> INSERT INTO personen VALUES (1, 'Skywalker', 'LUKE', NULL), (2, 'Solo', 'Ben', NULL);  
ERROR 1062 (23000): Duplicate entry '1' for key 'PRIMARY'  
mysql> SELECT * FROM personen;  
+-----+-----+-----+-----+  
| person_id | name      | vorname | geburtsdatum |  
+-----+-----+-----+-----+  
|          1 | Skywalker | Anakin  | NULL         |  
+-----+-----+-----+-----+  
1 row in set (0.00 sec)  
  
mysql> INSERT IGNORE INTO personen VALUES (1, 'Skywalker', 'LUKE', NULL), (2, 'Solo', 'Ben', NULL);  
Query OK, 1 row affected, 1 warning (0.00 sec)  
Records: 2  Duplicates: 1  Warnings: 1  
  
mysql> SHOW WARNINGS;  
+-----+-----+-----+  
| Level   | Code | Message                                     |  
+-----+-----+-----+  
| Warning | 1062 | Duplicate entry '1' for key 'PRIMARY'     |  
+-----+-----+-----+  
1 row in set (0.00 sec)
```

```
mysql> CREATE TABLE personen (person_id INT PRIMARY KEY AUTO_INCREMENT,  
-> name CHAR(35),  
-> vorname CHAR(35),  
-> geburtsdatum DATE);  
Query OK, 0 rows affected (0.05 sec)  
  
mysql> INSERT INTO personen VALUES (1, 'Skywalker', 'Anakin', NULL);  
Query OK, 1 row affected (0.01 sec)  
  
mysql> INSERT INTO personen VALUES (1, 'Skywalker', 'LUKE', NULL), (2, 'Solo', 'Ben', NULL);  
ERROR 1062 (23000): Duplicate entry '1' for key 'PRIMARY'  
mysql> SELECT * FROM personen;  
+-----+-----+-----+-----+  
| person_id | name       | vorname | geburtsdatum |  
+-----+-----+-----+-----+  
|          1 | Skywalker | Anakin  | NULL         |  
+-----+-----+-----+-----+  
1 row in set (0.00 sec)  
  
mysql> INSERT IGNORE INTO personen VALUES (1, 'Skywalker', 'LUKE', NULL), (2, 'Solo', 'Ben', NULL);  
Query OK, 1 row affected, 1 warning (0.00 sec)  
Records: 2  Duplicates: 1  Warnings: 1  
  
mysql> SHOW WARNINGS;  
+-----+-----+-----+  
| Level   | Code | Message                                     |  
+-----+-----+-----+  
| Warning | 1062 | Duplicate entry '1' for key 'PRIMARY'     |  
+-----+-----+-----+  
1 row in set (0.00 sec)  
  
mysql> SELECT * FROM personen;  
+-----+-----+-----+-----+  
| person_id | name       | vorname | geburtsdatum |  
+-----+-----+-----+-----+  
|          1 | Skywalker | Anakin  | NULL         |  
|          2 | Solo     | Ben     | NULL         |  
+-----+-----+-----+-----+  
2 rows in set (0.00 sec)
```



Daten einfügen: INSERT

```
INSERT [IGNORE]
  INTO tbl_name [(col_name,...)]
  VALUES ({expr | DEFAULT},...), (...), ...
  [ON DUPLICATE KEY UPDATE col_name = expr, ... ]
```

- **ON DUPLICATE KEY UPDATE** definiert das Verhalten der Einfügeoperation, wenn ein Schlüssel bereits in einer Tabelle enthalten ist
- Bei doppeltem Schlüsselwert wird der betreffende Tabelleneintrag mit den Vorgabewerten **col_name = expr** aktualisiert


```
mysql> CREATE TABLE personen (  
-> person_id INT PRIMARY KEY AUTO_INCREMENT,  
-> name CHAR(35),  
-> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE TABLE personen (  
  -> person_id INT PRIMARY KEY AUTO_INCREMENT,  
  -> name CHAR(35),  
  -> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> INSERT INTO personen VALUES (1,'Skywalker','Anakin');  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> CREATE TABLE personen (  
-> person_id INT PRIMARY KEY AUTO_INCREMENT,  
-> name CHAR(35),  
-> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> INSERT INTO personen VALUES (1,'Skywalker','Anakin');  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO personen VALUES (1,'Solo','Ben');  
ERROR 1062 (23000): Duplicate entry '1' for key 'PRIMARY'
```

```
mysql> CREATE TABLE personen (  
-> person_id INT PRIMARY KEY AUTO_INCREMENT,  
-> name CHAR(35),  
-> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> INSERT INTO personen VALUES (1,'Skywalker','Anakin');  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO personen VALUES (1,'Solo','Ben');  
ERROR 1062 (23000): Duplicate entry '1' for key 'PRIMARY'  
mysql> INSERT INTO personen VALUES (1,'Solo','Ben') ON DUPLICATE KEY UPDATE name='Solo', vorname='Ben';  
Query OK, 2 rows affected (0.00 sec)
```

```
mysql> CREATE TABLE personen (  
  -> person_id INT PRIMARY KEY AUTO_INCREMENT,  
  -> name CHAR(35),  
  -> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> INSERT INTO personen VALUES (1,'Skywalker','Anakin');  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO personen VALUES (1,'Solo','Ben');  
ERROR 1062 (23000): Duplicate entry '1' for key 'PRIMARY'  
mysql> INSERT INTO personen VALUES (1,'Solo','Ben') ON DUPLICATE KEY UPDATE name='Solo', vorname='Ben';  
Query OK, 2 rows affected (0.00 sec)  
  
mysql> SELECT * FROM personen;  
+-----+-----+-----+  
| person_id | name | vorname |  
+-----+-----+-----+  
|          1 | Solo | Ben      |  
+-----+-----+-----+  
1 row in set (0.00 sec)
```

```
mysql> CREATE TABLE personen (  
  -> person_id INT PRIMARY KEY AUTO_INCREMENT,  
  -> name CHAR(35),  
  -> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> INSERT INTO personen VALUES (1,'Skywalker','Anakin');  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO personen VALUES (1,'Solo','Ben');  
ERROR 1062 (23000): Duplicate entry '1' for key 'PRIMARY'  
mysql> INSERT INTO personen VALUES (1,'Solo','Ben') ON DUPLICATE KEY UPDATE name='Solo', vorname='Ben';  
Query OK, 2 rows affected (0.00 sec)  
  
mysql> SELECT * FROM personen;  
+-----+-----+-----+  
| person_id | name | vorname |  
+-----+-----+-----+  
|          1 | Solo | Ben      |  
+-----+-----+-----+  
1 row in set (0.00 sec)  
  
mysql> INSERT INTO personen VALUES (1,'Skywalker','Anakin') ON DUPLICATE KEY UPDATE name=NULL, vorname='YODA';  
Query OK, 2 rows affected (0.00 sec)
```

```
mysql> CREATE TABLE personen (  
  -> person_id INT PRIMARY KEY AUTO_INCREMENT,  
  -> name CHAR(35),  
  -> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> INSERT INTO personen VALUES (1,'Skywalker','Anakin');  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO personen VALUES (1,'Solo','Ben');  
ERROR 1062 (23000): Duplicate entry '1' for key 'PRIMARY'  
mysql> INSERT INTO personen VALUES (1,'Solo','Ben') ON DUPLICATE KEY UPDATE name='Solo', vorname='Ben';  
Query OK, 2 rows affected (0.00 sec)  
  
mysql> SELECT * FROM personen;  
+-----+-----+-----+  
| person_id | name | vorname |  
+-----+-----+-----+  
|          1 | Solo | Ben      |  
+-----+-----+-----+  
1 row in set (0.00 sec)  
  
mysql> INSERT INTO personen VALUES (1,'Skywalker','Anakin') ON DUPLICATE KEY UPDATE name=NULL, vorname='YODA';  
Query OK, 2 rows affected (0.00 sec)  
  
mysql> SELECT * FROM personen;  
+-----+-----+-----+  
| person_id | name | vorname |  
+-----+-----+-----+  
|          1 | NULL | YODA     |  
+-----+-----+-----+  
1 row in set (0.00 sec)
```



Daten einfügen: INSERT mit SET-Anweisung

```
INSERT [IGNORE]
INTO tbl_name
SET col_name={expr | DEFAULT}, ...
[ON DUPLICATE KEY UPDATE col_name=expr, ... ]
```

- alternative Schreibweise für **INSERT INTO ..VALUES**
- Fügt Daten basierend auf einer Menge in eine Tabelle ein
- **Beispiel:**

```
INSERT INTO personen SET person_id=1, name='Skywalker', vorname='Anakin'
entspricht: INSERT INTO personen VALUES (1, 'Skywalker', 'Anakin');
```



```
mysql> CREATE TABLE personen (person_id INT PRIMARY KEY AUTO_INCREMENT,  
-> name CHAR(35),  
-> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE TABLE personen (person_id INT PRIMARY KEY AUTO_INCREMENT,  
-> name CHAR(35),  
-> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> INSERT INTO personen SET person_id=1, vorname='Anakin';  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> CREATE TABLE personen (person_id INT PRIMARY KEY AUTO_INCREMENT,  
-> name CHAR(35),  
-> vorname CHAR(35));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> INSERT INTO personen SET person_id=1, vorname='Anakin';  
Query OK, 1 row affected (0.00 sec)  
  
mysql> SELECT * FROM personen;  
+-----+-----+-----+  
| person_id | name | vorname |  
+-----+-----+-----+  
|          1 | NULL | Anakin  |  
+-----+-----+-----+  
1 row in set (0.00 sec)  
  
mysql>
```



Daten einfügen: INSERT mit SELECT-Anweisung

```
INSERT [IGNORE]
[INTO] tbl_name [(col_name,...)]
SELECT ...
[ON DUPLICATE KEY UPDATE col_name=expr, ... ]
```

- Ermöglicht das Einfügen von Daten basierend auf einem SELECT-Statement
- Beispiel:

```
INSERT INTO die_skywalkers SELECT * FROM personen WHERE name = 'Skywalker';
```

```
mysql> SELECT * FROM personen;
```

person_id	name	vorname
1	Skywalker	Anakin
2	Skywalker	Padmé
3	Skywalker	Luke
4	Skywalker	Mara Jade
5	Solo	Han
6	Solo	Leia Organa
7	Solo	Anakin
8	Solo	Jaina

```
8 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM personen;
+-----+-----+-----+
| person_id | name      | vorname |
+-----+-----+-----+
| 1          | Skywalker | Anakin  |
| 2          | Skywalker | Padmé   |
| 3          | Skywalker | Luke    |
| 4          | Skywalker | Mara Jade |
| 5          | Solo      | Han     |
| 6          | Solo      | Leia Organa |
| 7          | Solo      | Anakin  |
| 8          | Solo      | Jaina   |
+-----+-----+-----+
8 rows in set (0.00 sec)

mysql> CREATE TABLE die_skywalkers LIKE personen;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM personen;
```

person_id	name	vorname
1	Skywalker	Anakin
2	Skywalker	Padmé
3	Skywalker	Luke
4	Skywalker	Mara Jade
5	Solo	Han
6	Solo	Leia Organa
7	Solo	Anakin
8	Solo	Jaina

```
8 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE die_skywalkers LIKE personen;  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> INSERT INTO die_skywalkers SELECT * FROM personen WHERE name='Skywalker';  
Query OK, 4 rows affected (0.00 sec)  
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM personen;
```

person_id	name	vorname
1	Skywalker	Anakin
2	Skywalker	Padmé
3	Skywalker	Luke
4	Skywalker	Mara Jade
5	Solo	Han
6	Solo	Leia Organa
7	Solo	Anakin
8	Solo	Jaina

```
8 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE die_skywalkers LIKE personen;
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> INSERT INTO die_skywalkers SELECT * FROM personen WHERE name='Skywalker';
```

```
Query OK, 4 rows affected (0.00 sec)
```

```
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM die_skywalkers;
```

person_id	name	vorname
1	Skywalker	Anakin
2	Skywalker	Padmé
3	Skywalker	Luke
4	Skywalker	Mara Jade

```
4 rows in set (0.00 sec)
```

```
mysql>
```


VIELEN DANK FÜR IHRE AUFMERKSAMKEIT!